IGCSE · Cambridge (CIE) · Computer Science

1 hour 15 questions

Exam Questions

Number Systems

Computers & Binary / Number Systems / Converting Between Binary & Denary / Converting Between Hexadecimal & Denary / Converting Between Hexadecimal & Binary / Uses of Hexadecimal / Binary Addition / Binary Shifts / Two's Complement

Easy (8 questions)	/43
Medium (6 questions)	/18
Hard (1 question)	/6
Total Marks	**/67**

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Easy Questions

1 (a) Binary is a number system used by computers.

Four 8-bit binary values are given.

Tick (✓) one box to show which 8-bit binary value is the correct conversion for the **denary value** 50.

A. 00101010

B. 00110010

C. 01001100

D. 01010000

(1 mark)

(b) Four 8-bit binary values are given.

Tick (✓) one box to show which 8-bit binary value is the correct conversion for the **hexadecimal value** 90.

A. 00001001

B. 01011010

C. 10010000

D. 01100100

(1 mark)

(c) **Explain** why a computer system can only process data in binary form

....

.... **(2 marks)**

(d) Two 8-bit binary values are given.

Add the two 8-bit binary values.

Give your answer **in binary**. Show all your **working**.

$$\begin{array}{r} 0\,0\,1\,1\,1\,0\,0\,1 \\ +\ 0\,1\,0\,0\,1\,0\,1\,0 \\ \hline \end{array}$$

(3 marks)

2 (a) All data needs to be converted to binary data so that it can be processed by a computer.

Explain why a computer can only process binary data.

....

.... **(2 marks)**

(b) The denary values 64, 101 and 242 are converted to 8-bit binary values.

Give the 8-bit binary value for each denary value.

64 ....

101 ....

242 ....

....

....

.... **(3 marks)**

(c) The hexadecimal values 42 and CE are converted to binary.

Give the binary value for each hexadecimal value.

42 ....

CE ....

....

....

....

.... **(4 marks)**

3 (a) Denary values are converted to binary values to be processed by a computer.

Draw one line from each denary value to the correctly converted 8-bit binary value.

Denary	8-bit binary
	00100001
41	10100110
	00101001
174	10000110
86	10101110
	01010110

(3 marks)

(b) Binary values can also be converted to denary values.

Give the correct denary value for the 12-bit binary value 000101010111

Show all your working.

....

.... **(2 marks)**

4 Hexadecimal is used for Hypertext Markup Language (HTML) colour codes.

An HTML colour code is:

#2F15D6

HTML colour codes and Media Access Control (MAC) addresses are two examples of where hexadecimal is used in Computer Science.

Give two other examples of where hexadecimal can be used in Computer Science.

....

.... **(2 marks)**

5 (a) A school network has several computers.

Each computer in the network has a media access control (MAC) address.

Hexadecimal is used for MAC addresses.

Part of a MAC address is given.

97–5C–E1

Each pair of digits is stored as binary in an 8-bit register.

Complete the binary register for these two pairs of digits.

97								

5C								

....

....

(4 marks)

(b) **Give two** other uses of hexadecimal in Computer Science.

(2 marks)

(c) Another value is stored as binary in a register.

0	1	0	1	0	0	1	0

A **logical left shift** of **two places** is performed on the binary value.

Complete the binary register to show its contents after this logical left shift.

(1 mark)

(d) **State one** effect this logical shift has on the binary value.

(1 mark)

6 (a) Binary is a number system used by computers.

Tick (✓) one box to show which statement about the binary number system is correct.

A. It is a base 1 system

B. It is a base 2 system

C. It is a base 10 system

D. It is a base 16 system

(1 mark)

(b) Denary numbers are converted to binary numbers to be processed by a computer.

Convert these three **denary numbers** to **8-bit binary numbers**.

50 ....

102 ....

221 ....

....

....

.... **(3 marks)**

(c) Two 8-bit binary numbers are given.

Add the two 8-bit binary numbers using binary addition.

Give your **answer in binary**. Show all your **working**.

```
 0 0 1 1 0 0 1 1
+0 1 1 0 0 0 0 1
----------------
```

....

....

.... **(3 marks)**

7 Hypertext markup language (HTML) colour codes can be represented as hexadecimal.

Tick (✓) one box to show which statement about the hexadecimal number system is incorrect

A. It uses the values 0 to 9 and A to F.

B. It can be used as a shorter representation of binary

C. It is a base 10 system.

D. It can be used to represent error codes.

(1 mark)

8 (a) The binary number 10100011 is stored in random access memory (RAM).

A logical left shift of three places is performed on the binary number.

Give the 8-bit binary number that will be stored after the shift has taken place.

.... **(1 mark)**

(b) **Tick (✓) one** box to show which statement about a logical left shift of **two** places is correct.

A. It would divide the binary number by 2.

B. It would multiply the binary number by 2.

C. It would divide the binary number by 4

D. It would multiply the binary number by 4.

(1 mark)

(c) 10100011 can be stored as a two's complement integer.

Convert the two's complement integer 10100011 to denary.

Show **all** your working.

....

.... **(2 marks)**

Medium Questions

1 Negative **denary** numbers can also be represented as binary using **two's complement**.

Complete the binary register for the denary value **−54**.

You must show all your **working**.

Register:								

(2 marks)

2 Binary is a number system used by computers.

Two 8-bit binary values are added.

The result of this calculation needs to be stored in an 8-bit register.

The denary result of this calculation is 301.

This generates an error

State the name of this type of error and **explain** why this error occurs.

(3 marks)

3 Two binary numbers are added by a computer and an overflow error occurs.

Explain why the overflow error occurred.

(2 marks)

4 Hexadecimal is used for Hypertext Markup Language (HTML) colour codes.

An HTML colour code is:

#2F15D6

Each pair of digits is stored as binary in an 8-bit register.

Give the 8-bit binary value that would be stored for each pair of hexadecimal digits.

- **2F**
- **15**
- **D6**

(6 marks)

5 Binary numbers are stored in registers.

Negative denary numbers can be represented as binary using two's complement.

Complete the binary register for the **denary** number **–78**

You **must** show all your **working**.

(2 marks)

6 Denary numbers can be converted to hexadecimal.

Convert the **three** denary numbers to hexadecimal

- **20**
- **32**
- **165**

....

....

.... **(3 marks)**

Hard Questions

1 Ron is attending a music concert. He has bought three tickets.

Each ticket number is displayed as a hexadecimal number.

Complete the table to show the **12-bit binary** values and the **denary** values for each Hexadecimal ticket number

Hexadecimal ticket number	12-bit binary value	Denary value
028		
1A9		
20C		

(6 marks)